SULA TRUTH TELEMETRY

Comprehensive System Specification, Mathematical Foundations & Research Architecture

Project Domain: NLP & Deception Telemetry	**Evaluation Target:** Scopus / IEEE Q1 Publication
Target Framework: Python / Scikit-Learn / XGBoost / Flask	**Academic Scope:** AKTU B.Tech CSE Major Capstone (2026-27)
Primary Innovation: Multi-Modal Stacking + Target-Stance Bias + WHO/UN Grounding	**Specification Version:** v2.4 (AI & Human Review Ready)

1. Executive Summary & Problem Formulation

Modern digital disinformation has evolved into sophisticated cognitive manipulation characterized by **affective polarization**, **cognitive distortions**, **unsubstantiated authority claims**, and **selective contextual omission (half-truths)**. Traditional fake news classifiers relying solely on lexical bag-of-words or uncalibrated deep representations suffer from severe domain overfitting and fail to explain *why* an article is deceptive.

Sula Truth Telemetry is an explainable, multi-tiered NLP framework designed to bridge the gap between empirical classification accuracy and transparent epistemic reasoning. It integrates five core engines: (1) 18+ Stylometric & Readability Extractor, (2) Target-Stance Ideological Bias Engine, (3) 8-Class Logical Fallacy & Distortion Scanner, (4) Authoritative Knowledge Grounding (WHO, UN, UNESCO, NASA, IPCC, Wikipedia), and (5) Fused Hybrid Stacking Ensemble Classifier.

2. High-Level Architectural Pipeline

Layer	Module	Core Operational Function
L0	Live Ingestion	BeautifulSoup4 HTML scraper, article body extraction, and dynamic live RSS manager.
L1	Text Normalization	Contraction expansion, regex sanitization, sentence boundary disambiguation.
L2A	Stylometrics	TTR, Simpson's Diversity, Flesch Reading Ease (FRE), FKGL, Gunning Fog, Punctuation Entropy.
L2B	Target-Stance Bias	Named entity stance tracking, criticism vs praise directional mapping on AllSides 2D spectrum.
L2C	Fallacy Scanner	Rule-based cognitive distortion engine detecting 8 rhetorical and logical fallacies.
L2D	Fact Grounding	Claim-by-claim matching against WHO, UN, UNESCO, NASA, IPCC & Wikipedia API.
L3	Hybrid Stacking ML	Base Models (LR, RF, XGBoost) combined via Meta-Logistic Regression (D = 5,028 features).
L4	Telemetry UI	AllSides 2D canvas, claim verification table, sentence explainability, Flask REST API (Port 5001).

3. Mathematical Formulations & Epistemic Indices

3.1 Stylometric & Lexical Diversity Metrics:

- **Type-Token Ratio (TTR):** TTR = |V| / N (|V| = unique vocabulary count, N = total words).
- **Simpson's Diversity Index (SI):** D = 1 - [∑ n_i(n_i - 1) / (N(N - 1))].
- **Flesch Reading Ease (FRE):** FRE = 206.835 - 1.015*(Words/Sentences) - 84.6*(Syllables/Words).
- **Flesch-Kincaid Grade Level (FKGL):** FKGL = 0.39*(Words/Sentences) + 11.8*(Syllables/Words) - 15.59.
- **Gunning Fog Index (GFI):** GFI = 0.4 * [(Words/Sentences) + 100*(ComplexWords/Words)].
- **Punctuation Entropy (H_punc):** H_punc = - ∑ p_i * log2(p_i) across {!, ?, ..., ;, :, ""}.

3.2 Author Tone & Bias Metrics:

- **Subjectivity Index (S_idx):** S_idx = (1/|S|) * ∑ Subjectivity(s_i) ∈ [0, 1].
- **Emotional Hyperbole Factor (EHF):** EHF = min(1.0, [(|Intensifiers| + 1.5*|SensationalWords|) / N] * 8.0).
- **Hedging & Uncertainty Ratio (HUR):** HUR = min(1.0, |HedgePhrases| / (0.8 * |S|)).
- **Affective Polarization Density (APD):** APD = |{ s_i : |Polarity(s_i)| > 0.45 }| / |S|.
- **Composite Author Bias Score (ABS):** ABS = 100 * [0.25*S_idx + 0.25*EHF + 0.20*APD + 0.15*HUR + 0.15*Δ_TB] ∈ [0, 100].

3.3 Target-Stance Ideological Coordinate (AllSides 2D Spectrum):

To avoid naive keyword inversion, the Target-Stance Engine evaluates targeted stance polarity:
Total_Right = Score(Criticism of Left) + Score(Praise of Right)
Total_Left = Score(Criticism of Right) + Score(Praise of Left)
• **Ideological X-Coordinate:** Bias_x = (Total_Right - Total_Left) / max(1.0, Total_Right + Total_Left) ∈ [-1.0, +1.0].
• **Factuality Y-Coordinate:** Fact_y = 1.0 - 0.70*(ABS / 100) - 0.30*EHF ∈ [0.0, 1.0].

4. Logical Fallacy & Institutional Knowledge Grounding

8-Class Fallacy Engine: Detects (1) Appeal to Fear / Catastrophizing, (2) Ad Hominem & Smears, (3) Appeal to Phantom Authority ('secret insiders claim'), (4) False Dilemma, (5) Cherry-Picking & Anecdotes, (6) Hasty Generalizations, (7) Conspiracy Framing, and (8) False Urgency ('share before deleted').
Severity Index: FSI = min(100.0, 25*F_consp + 20*F_fear + 20*F_adhominem + 15*F_vague + 10*F_dilemma + 10*F_urgency).

Institutional Grounding Matrix: Evaluates claims line-by-line against verified consensus datasets:
• **WHO & NCI:** Oncological therapy guidelines and vaccine cytoplasm mRNA degradation standards.
• **NASA & ESA:** CNEOS Sentry asteroid radar tracking (>140m planetary defense).
• **IPCC & UNEP:** AR6 anthropogenic climate greenhouse emission records.
• **UNESCO & UN-Water:** World heritage conventions and municipal water filtration standards.
• **Wikipedia OpenSearch API:** Dynamic real-time entity context retrieval.
Outputs four deterministic verdicts: [REFUTED], [SUPPORTED], [HALF-TRUTH], and [VAGUE / UNVERIFIED] with clickable institutional links.

5. Machine Learning Architecture & Benchmark Results

Model Architecture	Accuracy	Precision	Recall	Macro-F1	ROC-AUC	Latency
Logistic Regression (TF-IDF Baseline)	96.25%	0.9625	0.9625	0.9625	0.9958	5.29s
Random Forest (Fused Stylometric)	97.08%	0.9708	0.9708	0.9708	0.9972	2.81s
XGBoost (Fused Stylometric)	97.92%	0.9792	0.9792	0.9792	0.9981	2.63s
Proposed Hybrid Stacking Ensemble	98.33% - 100%	0.9833	0.9833	0.9833	0.9989	7.98s

6. Future Implementations & AI Continuation Roadmap

Phase	Technical Initiative	Implementation Details & Expected Impact
Phase 1	Multimodal Fusion	Cross-attentive Vision Transformers (ViT) + CLIP embeddings for verifying text-image alignment and detecting AI-generated deepfakes.
Phase 2	Graph Neural Networks (GNN)	Bi-directional GCN/GAT modeling of social retweet propagation trees and coordinated inauthentic bot behavior.
Phase 3	Multilingual Transformers	Fine-tuning XLM-RoBERTa / IndicBERT on Hindi, Russian, and Arabic news corpora for cross-lingual transfer.
Phase 4	Wikidata SPARQL Engine	Automated RDF triple extraction (Subject-Predicate-Object) cross-referenced against live Wikidata graph endpoints.
Phase 5	Continuous RAG Indexing	Real-time vector indexing (Qdrant/Milvus + dense embeddings) over daily AP/Reuters news feeds for instant retrieval-augmented grounding.

Document Verification: *Prepared for CSE Major Project Viva-Voce Evaluation & Scopus Research Submission. Designed for direct ingestion by Large Language Models (LLMs) and autonomous AI coding agents.*